

I N D E X



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Class 12 Section SA E Roll No. 31 Year 2022-23

Subject COMPUTER SCIENCE

Sl. No.	Experiment Description	Experiment Date	Submission Date	Remarks Signature
1.	To display the perfect square numbers from a list of integers.	15/5/22		
2.	To accept a sentence from the user and display the frequency of a substring in it. The substring can be a whole word or part of the word.	15/5/22		
3.	To convert a number entered by the user into ^{its} corresponding words using dictionaries.	15/5/22		
4.	To accept two tuples from the user and remove the elements from the second tuple which is present in the first tuple.	15/5/22		
5.	To calculate the volume of a box using defined function. The function should have appropriate default values for its parameters.	15/5/22		
6.	To accept a sentence, a character to be replaced, and ^a character to be replaced with, from the user.	16/5/22		

Sl. No.	Experiment Description	Experiment Date	Submission Date	Remarks Signature
7.	To write definition of a method <code>AFIND(cities)</code> to display all the city names from a list of <code>CITIES</code> , which are starting with 'A'.	16/5/22		
8.	To accept a list of numbers and display only the magic numbers from it using a user defined function. Magic N (list) passing the list as a parameter to the func.	17/5/22		
9.	A text file contains alphanumeric text (pocm.txt). To read this text file and print only those words that contain digits from the file.	28/7/22		
10.	To write a program to count total no. of "the" word present in a text file named "nice.txt".	28/7/22		
11.	Write a program that copies a text file "source.txt" into "target.txt" having the lines starting with a 'T'.	28/7/22		
12.	Writing a function <code>CreateFile()</code> to input n number of records in a binary file "Book.dat" with a structure <code>[Book No, BookName, Author, Price]</code> , where n is given by the user.	29/7/22		
13.	A func. <code>Counting()</code> that accepts an ^{argument} price. The func. should print the total no. of books that costs more than the value of the argument passed. We used the file <code>Book.dat</code> .	29/7/22		

Name Rajveer SinghClass 12 Section SC C Roll No. 31 Year 2021-22Subject COMPUTER SCIENCE

Sl. No.	Experiment Description	Experiment Date	Submission Date	Remarks Signature
14.	Writing a func. Update book() that will update the file 'Book.dat' created in the previous program. The function will change cost only. The new cost of the book will be given by the user.	27/7/22		
15.	Writing a func. filter_book() that will copy the information about the book by the given Author from 'Book.dat' to another file 'Booknew.dat'. After copying we delete Book.dat. Name of the author is passed as an argument.	29/7/22		
16.	We create a CSV file with a number of records and structure of each record in the format [id, destination, days, fare]. N is provided by the user. We write another function that will read and print the records in the file.	30/7/22		
17.	A func. that reads the before created CSV file and prints the records where the destination is less than 15 days. the function accepts the name of the file as an argument.	30/7/22		
18.	A CSV file with same structure as given in program 17. and having tab as delimiter. We also read and print the records in the file.	30/7/22		

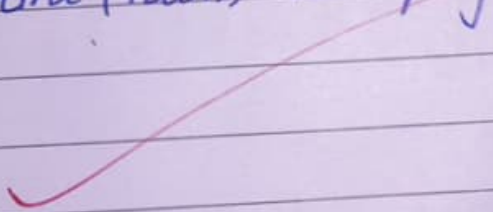
Sl. No.	Experiment Description	Experiment Date	Submission Date	Remarks Signature
19	To write a menu based program to add, peek, delete, and display the Record of LIBRARY using list as stack data structure in python. Record of LIBRARY contains the information BOOKID, BOOKTYPE, and BOOKCOST.	20/9/22		
20	Creating a dictionary containing names and marks as keys and value pairs of 6 students. To write a program to write separate user defined functions to perform the following operation (a) Push the keys of the dictionary into the stack where corresponding value (marks) > 75. (b) Pop (c) Displaying the content of the stack.	20/9/22		
21)	SQL Table :- 1	21/9/22		
22)	SQL Table :- 2	21/9/22		
23)	SQL Table :- 3	21/9/22		
24)	SQL Table :- 4	21/9/22		
25)	SQL Table :- 5	21/9/22		
26)	SQL Connectivity - 1	12/12/22		
27)	SQL Connectivity - 2	12/12/22		
28)	SQL Connectivity - 3	30/12/22		
29)	SQL Connectivity - 4	30/12/22		

PROGRAM 1

Q1) Write a program to display the perfect square numbers from a list of integers.

Ans 1)

```
a = int(input("Enter first integer"))  
b = int(input("Enter second integer"))  
for num in range(a, b):  
    for i in range(2, num):  
        if i * i == num:  
            print(num, "is a perfect square in the given list")
```



PROGRAM 2.

Q2) Write a program to accept a sentence from the user and display the frequency of a substring in it. The substring can be a whole word or the part of the word.

Ans 2)

```
a = input("Enter a sentence: ")  
b = input("Enter a substring: ")  
result = a.count(b)  
print(result)
```

PROGRAM 3.

Q3) Write a program using a dictionary to convert a number entered by the user into its corresponding number in words.

For example, if the input is 876 then the output should be 'Eighty Seven Six'.

Ans 3)

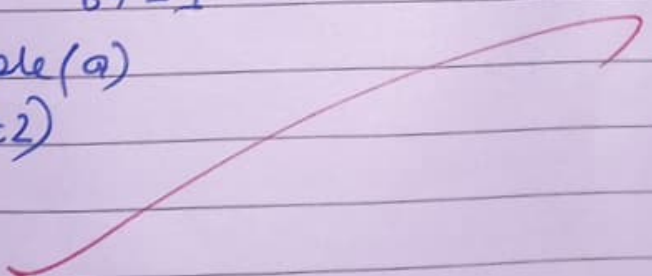
```
a = input("Enter a number: ")  
dict = {"0": "Zero", "1": "One", "2": "Two", "3": "Three",  
        "4": "Four", "5": "Five", "6": "Six", "7": "Seven",  
        "8": "Eight", "9": "Nine"}  
for num in dict:  
    print(dict[num], end = " ")
```

PROGRAM 4

Q4) Write a program to accept two tuples from the user and remove the elements from the second tuple which is present in the first tuple.

Ans 4) ~~t1 = tuple~~

```
t1 = tuple(input("Enter first tuple:"))
t2 = tuple(input("Enter second tuple:"))
a = list(t2)
b = 0
while b < len(t1):
    if t1[b] in a:
        a.remove(t1[b])
    else:
        b += 1
t2 = tuple(a)
print(t2)
```



PROGRAM 5

Q5) Write a program to calculate the volume of a box using defined function. The function should have appropriate default values for its parameters. The function should have the 3 input parameters - length of the box, width of the box, height of the box.

Ans 5)

```
def vol(a=1, b=1, c=1):  
    return a*b*c
```

```
L = int(input("Enter the length: "))
```

```
w = int(input("Enter the width: "))
```

```
h = int(input("Enter the height: "))
```

```
print("Volume of the box is = ", vol(L, w, h))
```

PROGRAM 6

Q6) Write a program to accept a sentence, a character to be replaced and a character to be replaced with, from the user.

A function `Replc(<str1>, <chr1>, <chr2>)` should have the mentioned parameters and will replace every occurrence of `<chr1>` with `<chr2>` in `<str1>`.

For eg: `str1 = "TECHNOLOGY"`

`chr1 = "O"`

`chr2 = "A"`

Output: `"TECHNALAGY"`

Ans 6) `def Replc(str1, chr1, chr2):`

`a = ""`

`for i in str1:`

`if i == chr1:`

`a += chr2`

`else:`

`a += i`

`print(a)`

`str1 = input("Enter a sentence: ")`

`chr1 = input("Enter the character to be replaced: ")`

`chr2 = input("Enter the character to be replaced with: ")`

`Replc(str1, chr1, chr2)`

PROGRAM 7

Q7) Write definition of a Method AFIND (CITIES) to display all the city names from a list of CITIES, which are starting with alphabet A.

For example: If the list CITIES contains

["AHMEDABAD", "CHENNAI", "NEW DELHI", "AMRITSAR", "AGRRA"]

The following should get displayed

AHMEDABAD

AMRITSAR

AGRRA

Ans) def AFIND(CITIES):

b = 0;

c = [];

for i in CITIES:

if (i[0] == 'A' or i[0] == 'a'):

c.append(i)

b = b + 1;

if (b == 0):

print("None of the cities in the given list start with a")

else:

print("Cities with start with 'A/a' are: ")

for b in c:

print(b)

List = eval(input("Enter a list of Cities: "))

AFIND(List)

Teacher's Signature

PROGRAM 8

Q8) Write a program to accept a list of numbers and display only the magic numbers from it using a user defined function MagicN(list) passing the list as a parameter to the function.

A magic number is a number which gives sum exactly

```
def sum(num):
```

Ans 8)

```
    str1 = str(num);
```

```
    num = 0
```

```
    for i in str1:
```

```
        i = int(i);
```

```
        num = num + i;
```

```
    return num;
```

```
def MagicN(list1):
```

```
    list = [];
```

```
    for z in list1:
```

```
        k = sum(z);
```

```
        while (len(str(k)) != 1):
```

```
            k = sum(k);
```

```
        if (k == 1):
```

```
            list.append(z);
```

```
    print("The magic numbers in this list are: ")
```

```
    for l in list:
```

```
        if (list.index(l) != (len(list) - 1)):
```

```
            print(l, end = ", ");
```

```
    else:
```

```
        print(l);
```

```
    nlist = eval(input("Enter a list of numbers: "));
```

```
    MagicN(nlist);
```

5/9

Text, using open()

PROGRAM 9

Q 9) A text file contains alphanumeric text(poem.txt).
Write a program that reads this text file and prints only those words that contain digits/number from the file.

Ans) `f = open("poem.txt", "r")`
`for i in f:`

`w = i.split()`

`for j in w:`

`for k in j:`

`if k.isdigit() and len(j) > 1:`

`print(j)`

`ctr += 1`

`if ctr == 1:`

`break`

`f.close()`

taking 1 line at a time

splitting that line into words

for each word in that line

for each letter in that word

Text, using with open()

Program 10

Q 10) Write a program in Python that counts total number of "the" word present in a text file named "nice.txt".

Ans) with open("nice.txt") as f:

c = 0

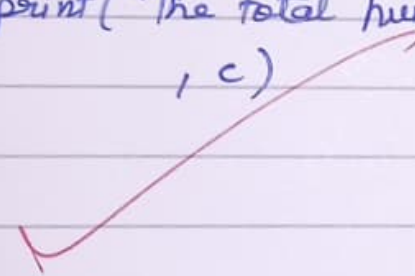
s = f.read()

for i in s.split():

if i.lower() == "the":

c = c + 1

print("The total number of 'the' word in this file is:"
 , c)



Text, using with open()

Program 11

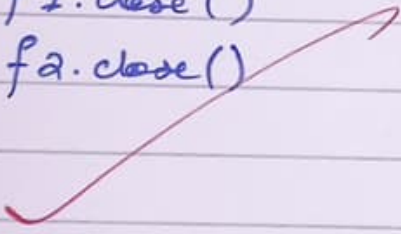
Q 11) Write a program that copies a text file "source.txt" onto "target.txt" basing the lines starting with a 'T'.

Ans 11) ~~f using~~

```
with open("source.txt", "r") as f1:  
    with open("target.txt", "w") as f2:  
        for i in f1:  
            if i[0] not in 'Tt':  
                f2.write(i)
```

f1.close()

f2.close()



Binary

Program 12

Qn) Write a function CreateFile() to input n number of records in a binary file "Book.dat" with a structure [BookNo, BookName, Author, Price], where n is given by the user.

Ans) `import pickle`

```
def CreateFile():
```

```
    f=open("Book.dat","wb")
```

```
    for i in range(a):
```

```
        Bno = int(input("Enter the BookNo:"))
```

```
        Bna = input("Enter the BookName:")
```

```
        A = input("Enter Author's Name:")
```

```
        P = int(input("Enter Price:"))
```

```
        L = [Bno, Bna, A, P]
```

```
        pickle.dump(L, f)
```

```
        L = []
```

```
    f.close()
```

```
    a = int(input("Enter the number of records of books:"))
```

```
    CreateFile()
```

Binary

Program 13

Q 13) Write a function counting() that accepts an argument price. The function should print the total number of books that costs more than the value of the argument passed. Use the file Book.dat created in the previous program.

Ans) `import pickle`

`def counting(price):`

`f = open("Book.dat", "rb")`

`c = 0`

`try:`

`while True:`

`L = pickle.load(f)`

`if L[3] > price:`

`c = c + 1`

`print(L)`

`except EOFError:`

`f.close()`

`print("The number of books that
are costlier are :", c)`

`a = int(input("Enter Price"))`

`counting(a)`

Binary

Program 14

Q 14) Write a function named `Update_book()` that will update the file 'Book.dat' created in the previous program. The function will change cost only. The new cost of the book will be given by the user.

Ans) `import pickle`

```
def Update_book(Bno, P):
```

```
    l = []
```

```
    f = open("Book.dat", "rb")
```

```
    try:
```

```
        while True:
```

```
            t = pickle.load(f)
```

```
            if Bno != t[0]:
```

```
                l.append(t)
```

```
            else:
```

```
                t[3] = P
```

```
                l.append(t)
```

```
    except EOFError:
```

```
        f.close()
```

```
        s = open("Books.dat", "wb")
```

```
        for i in l:
```

```
            pickle.dump(i, s)
```

```
        s.close()
```

```
    a = int(input("Enter the Book No: "))
```

```
    b = int(input("Enter the new cost: "))
```

```
    Update_book(a, b)
```

Teacher's Signature

Binary.

Program 5

Q15) Write a function filter_book() that will copy the information about the book by the given Author from 'Book.dat' to another file 'Book_new.dat'. After copying delete Book.dat. The name of the author will be passed as an argument to the function.

Ans) import pickle
import os

```
def filter_book(authorname):
    f = open("Books.dat", "rb")
    s = open("Book_new.dat", "wb")
    try:
        while True:
            L = pickle.load(f)
            if L[2] == authorname:
                pickle.dump(L, s)
    except EOFError:
        f.close()
        s.close()
    os.remove("Books.dat")
    was
```

```
authorname = input("Enter author's name: ")
filter_book(authorname)
```

Teacher's Signature

Program 16

Q16) Create a file (csv) with n number of records and structure of each record is in the format [id, destination, days, fare]. N is provided by the user. Write another function that will read and print the records in the file.

Ans)

```

import csv
def Reader():
    f = open("travel.csv", "r")
    csvr = csv.reader(f)
    for i in csvr:
        print(i)
    f.close()

f = open("travel.csv", "w", newline = " ")
csvw = csv.writer(f)
n = int(input("Enter the number of records: "))
for i in range(n):
    L = []
    iid = int(input("Enter the id number: "))
    dest = input("Enter destination: ")
    days = int(input("Enter the number of days: "))
    fare = int(input("Enter the fare: "))
    L = [iid, dest, days, fare]
    csvw.writerow(L)

f.close()
Reader()

```


CSV

Program 17

Q 17) Write a python function that reads the before created CSV file and prints the records where the destination takes ^{less} than 15 days. The function accepts the name of the file as argument.

```
import csv
def ReadRead(n):
    f=open(n,"r")
    csvr = csv.reader(f)
    c = 0
    for i in csvr:
        k = int(i[2])
        if k < 15:
            print(i)
            c += 1
    if c == 0:
        print("No travel record within 15 days!")
    f.close()
```

n = input("Enter the name of the file:")
Read(n)

CSV

Program 18

Q 18) Write a CSV file with same structure as given in the previous question (program 17) and having tab as delimiter. Also, read and print the records in the file.

```
import csv
def Reader():
    f = open("travel.csv", "r")
    csvr = csv.reader(f, delimiter = '\t')
    for i in csvr:
        print(i)
    f.close()
```

```
f = open("travel.csv", "w", newline = " ")
csvw = csv.writer(f, delimiter = '\t')
n = int(input("Enter the number of records: "))
for i in range(n):
    l = []
    iid = int(input("Enter the id no: "))
    dest = input("Enter the destination: ")
    days = int(input("Enter the number of days: "))
    fare = int(input("Enter the fare: "))
    l = [iid, dest, days, fare]
    csvw.writerow(l)
f.close()
Reader()
```

AD
22/8/22

PROGRAM 19

Q) Write a menu based program to add, peek, delete and display the record of LIBRARY using list as stack data structure in python. Record of LIBRARY contain the information Book ID, BookType, and BookCost.

→ `stack = []`

`def Add(item):`

`stack.append(item)`

`top = len(stack) - 1 // 0`

`def Pop():`

`if stack == []:`

`top = None - 1`

`return "Underflow"`

`else:`

`item = stack.pop() // 3`

`if len(stack) == 0:`

`top = None`

`else:`

`top = len(stack) - 1. 2 - 1 = 2. 1`

`return stack[top]`

`def Display():`

`if stack == []:`

`print("Stack is empty")`

`else:`

`print("The Stack elements are:")`

`top = len(stack) - 1 // 3 - 1 = 2`

`for i in range(top, -1, -1):`

`print(stack[i])`

top	3
top-1	2
0-top	10

top=0	1
top=-1	2

Teacher's Signature

while True:

print("Menu")

print("1. Add")

print("2. Pop")

print("3. Peek")

print("4. Display")

print("5. Exit")

ch = int(input("Enter your choice:"))

if ch == 1:

Bookid = int(input("Enter book id:"))

Booktype = input("Enter book type:")

Bookcost = int(input("Enter the cost of book:"))

L1 = [Bookid, Booktype, Bookcost]

Add(L1) st [[], []]

elif ch == 2:

item = Pop()

if item == "underflow":

print("Unable to delete")

else:

print("Deleted item:", item)

elif ch == 3:

item = Peek()

if item == "underflow":

print("Empty Stack")

else: print("Top Element:", item)

elif ch == 4:

Display()

elif ch == 5:

break

else:

print("Error")

Teacher's Signature

PROGRAM 20

Q) Create a dictionary containing names and marks as key value pairs of 6 students. Write a program, with separate user defined functions to perform the following operations:

a) Push the Keys of the dictionary into the stack, where corresponding value (marks) > 75.

b) Pop

c) Display the content of the stack.

```
def Push():
```

```
    for Key in d:
```

```
        if d[Key] > 75:
```

```
            stack.append(Key)
```

```
def Pop():
```

```
    if not stack:
```

```
        return "underflow"
```

```
    else:
```

```
        return stack.pop()
```

```
def Display():
```

```
    if not stack:
```

```
        print("Empty Stack")
```

```
    else:
```

```
        print("The elements are:")
```

```
        top = len(stack)-1
```

```
        for i in range(top, -1, -1):
```

```
            print(stack[i])
```


SQL-TABLE 1

I)

relation Student

No	Name	Age	Department	Date of adm	Fee	Gender
1.	Pankaj	24	Computer	10/01/97	120	m
2.	Shalini	21	History	24/03/98	200	f
3.	Sanjay	22	Hindi	12/12/96	300	m
4.	Sudha	25	History	01/07/99	400	f
5.	Rakesh	22	Hindi	05/09/97	250	m
6.	Shakeel	30	History	27/06/98	300	m
7.	Swiya	34	Computer	25/02/97	210	m
8.	Shikha	23	Hindi	31/07/97	200	f

Q1) To show information of all students of History Department.

→ select * from Student where Department = "History";

Q2) To list the names of female students who are in Hindi Department.

→ select Name from Student where Gender = "f" and Department = "Hindi";

Q3) To display the names of all students when their date of admission in ascending order.

→ select Name, Dateofadm from Student order by Dateofadm asc;

Q4) To count the number of students who were born in 1998.

→ select count(*) as Number of Students from Student where Dateofadm like "1998-%";

Q5) To display the names of ~~the~~ all departments.

→ Select distinct Department from Student;

SQL - TABLE 2

II)

CARDEN

Ccode	Car Name	Make	Color	Capacity	Charges
501	A-Star	Suzuki	RED	3	14
503	Indigo	Tata	SILVER	3	12
502	Innova	Toyota	WHITE	7	15
509	SX4	Suzuki	SILVER	4	14
510	C Class	Mercedes	RED	4	35

Q1) Create the above table and insert all the records.

→ create table Carden (Ccode int, CarName varchar(10), Make varchar(10), Color varchar(10), Capacity int, Charges int);

insert into Carden values (501, "A-Star", "Suzuki", "RED", 3, 14),
 (503, "Indigo", "Tata", "SILVER", 3, 12),
 (502, "Innova", "Toyota", "WHITE", 7, 15),
 (509, "SX4", "Suzuki", "SILVER", 4, 14),
 (510, "C class", "Mercedes", "RED", 4, 35);

Q2) To display all the information about the Suzuki cars.

→ select * from CARDEN where Make = "Suzuki";

Q3) To display the Ccode, CarName, and color in descending order of their capacity.

→ select Ccode, CarName, Color from Carden order by Capacity desc;

Q4) To display the names of all the ~~red~~^{colored} cars that can accommodate 4 people.

→ select CarName from Carden where Color = "RED" and Capacity = 4;

Q5) To display the total number of models (make)

→ select Count (distinct make) from Carden;

Q6) To display the information of cars where the name starts with the letter 'I' in ascending order of CarName.

→ select * from Carden where CarName like "I%" order by CarName;

Q7) To display the names of cars where the capacity is more than 4 and Ccode is more than 505.

→ select CarName from Carden where Capacity > 4 and Ccode > 505;

Q8) To display no. of cars of each make.

→ select ^{distinct} Make, count (*) from Carden group by Make;

SQL-TABLE 3.

III)

Items

CODE	INAME	QTY	PRICE	COMPANY	TCODE
1001	DIGITAL PAD 12i	120	11000	XENITA	T01
1006	LED SCREEN 40	70	38000	SANTORA	T02
1004	CAR GPS SYSTEM	50	21500	GEO KNOW	T01
1003	DIGITAL CAMERA 12K	160	8000	DIGICLOCK	T02
1005	REN DRIVE 32GB	600	1200	STOREHOME	T03

TRADERS

TCODE	TNAME	CITY
T01	ELECTRONIC SALES	MUMBAI
T03	BUSY STORE CORP	DELHI
T02	DISP HOUSE INC	CHENNAI

Q1) Display the TNAME, TCODE, INAME from the tables.

→ select T.TNAME, I.TCODE, T.TCODE, I.INAME from
Items I, TRADERS T where I.TCODE = T.TCODE;

Q2) Display the INAME, TCODE, TNAME where quantity is less than 100.

A → select INAME, T.TCODE, T.TNAME from Items I,
TRADERS T where I.TCODE = T.TCODE and QTY < 100;

Q3) Display TNAME and the total number of items that each trader bought.

→ select TNAME, ^{sum(QTY)}
~~count(TNAME)~~ as Total-number-of-items-bought
from Items I, TRADERS T where I.TCODE = T.TCODE
group by TNAME;

Q4) Display the total number of companies each trader had bought.

→ select T.TNAME, ~~count~~ ^{count}(~~distinct~~ COMPANY) as Companies
from Items I, Traders T where I.TCODE = T.TCODE
Group by T.TNAME;

SQL - TABLE 4

IV)

SHOP

ID	SName	Area
S01	ABC Computronics	CP
S02	AM Infotech Media	GKII
S03	Tech Shoppe	CP
S04	Geeky Techno Soft	Nehru Place
S05	Hitech Store	Nehru Place

ACCESSORIES

IP	IName	Price	SNO
A01	Mother Board	12000	S01
A02	Hard Disk	5000	S01
A03	Keyboard	500	S02
A04	Mouse	300	S01
A05	Mother Board	13000	S02
A06	Keyboard	400	S03
A07	LCD	6000	S04
A08	LCD	5500	S05
A09	Mouse	350	S05
A10	Hardisk	4500	S03

Q1) To display Name and Price of the accessories in ascending order of its price.

→ select IName, Price from Accessories order by Price;

Q2) To display the ID and SName of all Shops located in Nehru Place.

→ select id, SNAME from Shop where area = "Nehru Place";

Q3) To display the minimum and maximum price of each name of accessories.

→ select IName, min(Price) as Min-price, max(Price) as Max-price from Accessories group by IName;

Q4) To display the Name, Price of all the accessories and its respective SNAME where they are available.

→ select IName, Price, SName from Shop S, Accessories A where S.ID = A.SNo;

Q5) Display the total number of items that each shop sells along with the shop names.

→ select SName, count(~~distinct~~ IName) as Number-of-items-sold from Shop S, Accessories A where S.ID = A.SNo group by SName;

SQL-TABLE 5

Q1)

Employee

Eid	Name	Dept Id	Qualification	Sex
1	Deepali Gupta	101	MCA	F
2	Rajat Tyagi	101	BCA	M
3	Hari Mohan	102	B.A.	M
4	Harry	102	M.A.	M
5	Sunit Mittal	103	B.Tech	M
6	Jyoti	101	M.Tech	F

Salary

Eid	Basic	P.A	HRA	Bonus
1	6000	2000	2300	200
2	2000	300	300	30
3	1000	300	300	40
4	1500	390	490	30
5	8000	900	400	80
6	10000	300	490	89

Q1) To display the frequency of employees department wise

→ Select Deptid, count(*) as Frequency from Employee group
by Deptid;

or

select Deptid, count(Name) as Frequency from Employee group
by Deptid;

Teacher's Signature

Q2) To list the names of those employees only whose name starts with 'H'.

→ select Name from Employee where Name like "H%";

Q3) To add a new column in salary table. The column name is Total_Sal.

→ alter table Salary add Total_Sal int;

Q4) To store the corresponding values in Total_Sal column.

→ Update Salary set Total_Sal = Basic + DA + HRA + Bonus;

Ans
9/11/22

Programme 26.

A table "Event" in the database "Programme" has the following fields and datatypes:

E.No $\rightarrow$ char(4)

E.Name $\rightarrow$ varchar(15)

NOP $\rightarrow$ int

E.Date $\rightarrow$ Date

Write a program to print the following $\rightarrow$

① To fetch only those records from the "Event" Table where NOP is more than 10.

② To display the table contents in ascending order of E.date.

```
import mysql.connector as mc
```

```
objectt = mc.connect(host='localhost', user='root',  
port=3307, password='1234', database='programme')
```

```
dbc = objectt.cursor()
```

```
p1 = dbc.execute("Select * from Event where NOP > 10")  
print (dbc.fetchall())
```

```
p2 = dbc.execute("select * from Event order by E.date")  
print (dbc.fetchall())
```

PROGRAM 21

A table "Employee" in the database "ABC Corps" has the following field and datatypes:

Emp. No $\rightarrow$ char(4)

Emp - Name $\rightarrow$ varchar(15)

Gender $\rightarrow$ char(1)

Salary $\rightarrow$ int

DOJ $\rightarrow$ date

Design a Python program to do the following:

(a) Accept an year from the user in the form "yyyy".

Display all "female" employees who joined the organisation that year.

(b) To display gender wise Maximum and minimum salary.

```
import mysql.connector as mc;
```

```
myconnect = mc.connect(host = "localhost", user = "root",  
                        passwd = "1234", database = "ABC Corps")
```

```
dbcursor = myconnect.cursor()
```

```
dbcursor.execute("Select * from Employees where  
gender = 'F' ")
```

```
data = dbcursor.fetchall()
```

```
yyear = int(input("Enter year to look for: "))
```


for x in data :

k = str(x[4])

n = k[0:4]

if (str(n) == str(year)):

print(x[0], x[1], x[2], x[3], x[4])

print()

print("Maximum and minimum are :")

dbcursor.execute("Select gender, max(salary),
min(salary), from Employee group by gender")

data2 = dbcursor.fetchall()

for m in data2:

print(m[0], m[1], m[2])

myconnect.close()

PROGRAM 28

A table "customer" in the database "prog" has the following fields and datatypes:

serial number $\rightarrow$ int

name $\rightarrow$ varchar (15)

address $\rightarrow$ varchar (60)

Write a program to print the address of those customers where the word 'way' is contained in the address field.

```
import mysql.connector
```

```
mydb = mysql.connector.connect (host = "localhost",  
                                user = "root", password = "1234", database = "prog")
```

```
cr = mydb.cursor()
```

```
sql = "SELECT * FROM customer WHERE  
      address like '%way%'"
```

```
cr.execute (sql)
```

```
result = cr.fetchall()
```

```
for x in result:
```

```
    print(x)
```

AG
31/1/23
X

PROGRAM 29

A table "customer" in the database "prog" has the following fields and datatypes:

serial number $\rightarrow$ int

name $\rightarrow$ varchar(15)

address $\rightarrow$ varchar(60)

Write a program to print the first 5 records from the table "customer".

```
import mysql.connector
```

```
db = mysql.connector.connect(host = "localhost";
```

```
user = "root", password = "1234", database = "prog")
```

```
cr = db.cursor()
```

```
cr.execute("SELECT * FROM customer LIMIT 5;")
```

```
result = cr.fetchall()
```

```
for x in result:
```

```
print(x)
```

X

AN
3/1/23